

Blockchain-Based IoT Device Authentication and Cost-Effective Key Management Framework

Geetha Manoharan
School of Business, SR University,
Warangal, Telangana, India
geethamanoharan1988@gmail.com

Potnuri Suribabu
Professor, Computer Science and
Engineering, KMM Engineering
College, JNTUA University, Tirupati,
AP.
suribabupotnuri786@gmail.com

B. Venugopal
Indian Institute of Plantation
Management Bengaluru, Jnana
Bharathi Campus, Malathalli,
Bengaluru
venugopal.b@iipmb.edu.in

Mohit Tiwari
CSE Department, Bharati Vidyapeeth's
College of Engineering, Delhi A-4,
Rohtak Road, Paschim Vihar, Delhi
mohit.tiwari@bharativedyapeeth.edu

Kannan Kunnathully
Assistant Professor,
Veltech School of law
kannankunnathully@gmail.com

Vandana Roy
Department of Electronics
Communication, Gyan Ganga Institute
of Technology and Sciences, Jabalpur,
M.P., India
vandana.roy20@gmail.com

Abstract — IoT is seen as an exciting innovation that might start an entirely novel industrial revolution. A collection of Internet-connected devices. An unavoidable and pervasive framework is still an open problem in IoT research for embedded devices, neural networks, machine learning, and other widespread and universal systems. IoT systems rely on sensors and controllers. IoT uses millions of sensors to gather and share outside information with the public internet. Many applications analyze and evaluate diverse data collected from devices. Those compact devices for sensors have limited processing power. Security is crucial for precise data flow between verified users, which becomes more complicated in a network of IoT devices. IoT's decentralized architecture design faces security concerns include privacy, reliability, security, anonymity, and device-user trust. Due to limited resources in sensor devices, typical cryptographic techniques for identification and organizational management of keys are not suitable for IoT systems. Effective protection against electronic card theft, session data publication, external attack, passwords guessing, user sensing simulation, and anonymity for users is provided by the suggested technique. Because it meets all security criteria, the updated protocol offers better security. The simplest protocol that can survive all known attacks and provide the essential security is the suggested one. With a 99% success rate and superior performance across all measures, the findings show that the suggested blockchain technology IoT identification framework is the way to go.

Keywords- Internet of Things, networks, security, cryptographic, protocol, privacy, sensors.

I. INTRODUCTION

Since IoT-based systems rely on centralized protocols for authentication and authorization, there is a potential for breaches of security due to the proliferation of these systems. This creates additional concerns about security and confidentiality. While registration, confidentiality, and authenticity are the primary issues of IoT security, there are

risk aspects such as unlawful authorization, database breaches, and theft of identities. In order to ensure that every node in the Internet of Things (IoT) network is legitimate, the process for authentication is crucial. It is common practice to wirelessly hack IoT devices in an effort to get access to them over a web connection [1]. Users' skill in securing the network's constituent devices determines the IoT network's viability. Prioritizing security is crucial when integrating IoT devices into corporate networks. Adopting robust yet efficient cryptographic methods is essential for standardizing safe communication between machines, which is crucial for solving this challenge.

Therefore, cybercriminals are going to target smart devices more often as their use increases. Adversaries may monitor gadgets and gain unauthorized usage of internet of things networks and gadgets, which might lead to illicit activities such as the disclosure of private data. Thus, when identification is compromised, even just one node could prove deadly [2-3]. On top of that, it either triggers disasters or completely turns off the system. The authentication method is used to confirm the identification of every endpoints in the IoT system. Upon signing up, the process for registration is set up and the service supplier is informed of the mechanism for confirming the system's identification. Therefore, to stop a malevolent device from connecting to an IoT network, an adaptable security strategy that is both affordable is required. So, it lets a device generate a new authenticating identity every time it joins a network.

The Internet of Things (IoT) has proliferated and found several uses in modern society, such as "smart" residences, "smart" towns, "smart" medical care, etc., leading to the connectivity of potentially hundreds of millions of devices. As a result, authenticating IoT devices in this kind of network is seen as an essential need for security of the system. The investigators have come up with a number of ideas on how to fix these issues. But because of their one-of-a-kind

characteristics, conventional authentication techniques are useless and inappropriate for Internet of Things devices.

Cryptographic approaches are not appropriate for the limited in resources nodes of the Internet of Things (IoT) due to their high calculation and cost of transmission. Because of their storage requirements and optimization for high-processor computers, cryptographic algorithms are notoriously resource intensive. By integrating the technology of blockchain, which shows great promise, into an ecosystem where Internet of Things nodes are already present, this problem may be resolved [4-6]. Given the nature and one-of-a-kind nature of interactions among IoT smart gadgets, the suggested solution incorporates blockchain architectural elements such as immutable ledgers and decentralized verification. The cryptographic aspect of the blockchain system makes it safer; additionally, it aids in the identification of identities and the authorization of resources for nodes within the Internet of Things (IoT) architecture. To sum up, current solutions do not address the issue of cost-effectively protecting IoT devices from illegal usage via a mutual authenticating strategy.

II. RELATED WORKS

A research was conducted with the purpose of establishing lightweight authentication method for Internet of Things devices by incorporating cryptographic techniques that are efficient with resources [7-8]. This method enables safe communication without placing an excessive load on devices that have limited processing capacity; nonetheless, it is only applicable in certain inexpensively situations.

The blockchain relies control of access is being suggested as a means of improving network safety for the Internet of Things (IoT). Although the employing of smart contracts guarantees decentralized and impermeable interactions, the implementation of a system of this kind may result in an upsurge in the level of complexity of the distributed system and the amount of resources that are consumed [9-10].

A system for managing confidence that is based on fuzzy logic and the calculation of dynamic trust ratings for Internet of Things devices was proposed by investigators [11-13]. However, periodic updates are necessary to maintain correct trust ratings in cloud-based settings, despite the fact that this approach helps making choices in such contexts.

To improve the security of session-based authorization, an OTP (one-time password) authentication method that has been optimized was created [14-15]. This method generates temporary encryption codes. It is important to note that while this method is capable of preventing unwanted access, it is only capable of verifying one instance and needs human

participation for every session. Through the process of breaking Internet of Things systems into small divisions, a distributed control of keys strategy was implemented in order to solve the issue of single-point failure [16]. This approach assures that interaction is safe, but it results in a significant amount of interaction overhead throughout the process of reconfiguration.

The identification of anomalies in Internet of Things settings was accomplished via the use of a structure for deep learning that incorporates both multilayer and recurrent neural networks for the purpose [17]. Applications that operate continuously may find it difficult to meet the computational requirements, despite the fact that it provides a high level of detection precision. An RFID system that is safe for use in distribution networks was developed by researchers who used blockchain technology to store records of product monitoring [18]. Despite the fact that this protects the reliability and permanence of monitoring data, adoption may be limited due to the costly nature of installation and storage needs.

Through the use of intelligent gadgets, an electronic gait identification system was built in order to identify people according to the patterns of their stepping [19]. Despite the fact that it offers ongoing authorization, its ability to perform may suffer if users change the speed or rhythm at which they walk.

In order to guarantee secure interaction between Internet of Things nodes, a decentralized key administration system that makes use of logical value hierarchy is required [20-23]. On the other hand, one of the most significant drawbacks of this technique is the substantial communication cost that is connected with the maintenance of the most important structure.

III. PROPOSED METHODOLOGY

Our proposed blockchain-based identification solution for IoT sensors aims to make them more trustworthy and secure. The idea beneath the blockchain-based solution is to use the distributed feature, which provides a new angle on identification and security for Internet of Things devices. The safety landscape has become more complex due to the rapid evolution of technologies. Beginning with basic hashing computations, it eventually developed sophisticated encrypted storing systems. They are saved and linked based on historical data by the cryptography nature of the blockchain framework as mentioned in the figure 1. The objective of this approach is to lessen the likelihood of any Internet of Things device being exploited by means of unauthorized access, including probing. As an archive and a method of keeping documents safe from other parties or visitors, blockchain is used in our recommended approach. This kind of security evolved from older methods of

encrypting it which had several drawbacks such as privacy issues and lengthy processing times. With its decentralized design, the suggested design overcomes the shortcomings of centralized approaches such as computing cost, susceptibility, and one source of failure.

sensing devices, cluster heads, network gateways, and cryptocurrency administrators are essential parts.

3.2 System Initialization Module:

As part of this portion, while production, Internet of Things (IoT) devices are initialized with an OID, publicly key, and secret key pair. When authenticating devices, the associated keys and IDs are crucial. Private keys are generated using the Elliptic Curve Digital Signature Algorithm (ECDSA), which guarantees a safe and compact key creation method.

$$SK = h(OID \parallel PUB) \quad (1)$$

A device's public key is denoted by PUB, its unique identification is represented by OID, and the secret key is denoted by SK.

3.3 Registration Module:

The administrator of the blockchain may verify the identity of every node sensors, gateways, and clusters heads by looking up their distinctive OID. The smart contract stored on the blockchain keeps a permanent record of all connected devices, thus this authentication can be easily tracked.

$$Device_{ID} = h(OID \parallel T_{join}) \quad (2)$$

The date and time whenever the gadget enters a network is denoted as T_{join} . This guarantees the safe tracking of each device's registrations throughout times.

3.4 Identification Module:

Each enrolled device gets an individual system device ID during the registration phase. This ID is a combination of the cluster head's ID, the nonce value, and the unique OID. In order for devices to communicate with each other on a shared network, this system of device ID makes sure that they are all individually identifiable.

$$System_{DeviceID} = h(OID \parallel CHID \parallel N1) \quad (3)$$

To avoid replay attacks, the cluster head's identification (CHID) and a nonce value (N1) are used.

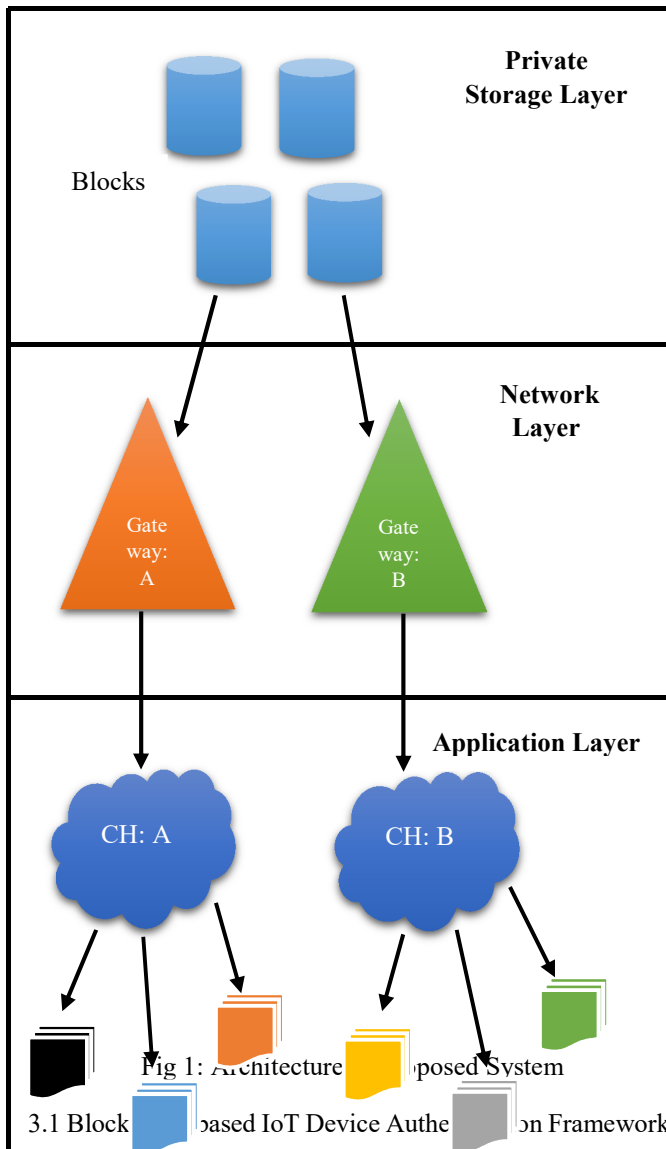
3.5 Authentication Module:

To prevent unauthorized devices from connecting to the the network, the authorization module checks their credentials. Gadgets communicate with the cluster head by sending authenticated join request. The requests are confirmed by their cluster leader via a verification of the blockchain data.

$$Join_{Request} = ESK(Device_{ID} \parallel OID \parallel T1 \parallel N1) \quad (4)$$

Devices cannot join a network unless they have the right secret key, which is protected by the encryption challenge-response system.

3.6 Secret Key Generation Module:



3.1 Blockchain-based IoT Device Authentication Framework

To improve the identification and safety of Internet of Things (IoT) devices, a Physical Devices approach using blockchain technology. The decentralization of identification

solves the problems with centralized systems, including elevated computing costs and single-point errors. Using smart contracting for automated identification and privilege management, the structure takes use of the blockchain's cryptography properties to provide safe, permanent documentation of devices interactions. Intelligent agreements,

Utilizing secret keys created for particular communication lines, this module creates encrypted channels for devices to communicate with one another.

- Gateway and Blockchain Admin:

$$SK_{GWB} = h(GWID \parallel PUB_{Admin}) \quad (5)$$

- Cluster Head and Gateway:

$$SK_{CHG} = h(CHID \parallel PUB_{GW}) \quad (6)$$

- Sensor Device and Cluster Head:

$$SK_{SDC} = h(SDID \parallel PUB_{CH}) \quad (7)$$

The corresponding entities are able to communicate securely and encrypted thanks to these keys.

3.7 Challenge-Response Authentication Module:

A problem is sent to the sensing device by the head of the cluster, which is encrypted with the secret key that corresponds to the task at hand. Decryption of the difficulty, computation of the needed values, and transmission of an answer are all performed by the sensing device.

$$Response = DSK \left(ESK \left(System_{DeviceID} \parallel OID \parallel T1 \parallel N1 \parallel N2 \right) \right) \quad (8)$$

The gadget is considered to be validated and its private keys are added to the blockchain if the entered values are verified to be same.

3.8 Zero-Knowledge Proof (ZKP) Integration Module:

The ZKP protocols are used by the proposed system in order to identify devices without divulging their private keys. The person who proves in ZKP is the one who demonstrates understanding of the secrets while divulging it to the validator in order to persuade them of the veracity of the prover's claim. This guarantees that both privacy and safety are maintained.

V. RESULT

In this part, we assess the suggested method by looking at the requirements for processing, interaction, and storing in order to verify the safekeeping of IoT devices with sensors. We may deduce whether the suggested system demonstrates greater safety after comparing the suggested scheme's effectiveness with existing protocols. The updated protocol offers enhanced protection as it satisfies all of the safety requirements. The only method that provides all required security characteristics while also being impervious to all known attacks is the one that has been presented.

A number of important characteristics, including as identification achievement rate, energy consumption, overhead for communication, and computational cost, were taken into consideration while evaluating the a blockchain based Internet of Things identification architecture that was

developed recently. These indicators are helpful in determining whether or not the structure is able to guarantee safe, cost-effective, as well as effective communications in Internet of Things contexts that are limited in resources. Through the utilization of intelligent contracts for safe identification and distributed storage, the network is able to lessen its reliance on resources provided by other parties, hence increasing its level of robustness. A comparison of the suggested structure with various alternatives is shown in the following two tables, which are accompanied by clarifications.

Authentication Success Rate (%): Indicates trustworthiness by quantifying the proportion that has achieved authorizations.

Energy Consumption (%): Determines efficiency by comparing energy use to the benchmark model.

Processing Cost (%): Specifies the computing effort required for authenticating devices, represented as an amount.

Communication Overhead (%): The proportion of overall transmission that consists of extra data transferred during verification.

Latency (%): Identifies the amount of lag time between identification and the starting point model.

Security Level (%): Finds the model's risk reduction effectiveness as percentage.

Table 1: Performance Evaluation of similar methods

Technique	Authentication Success Rate (%)	Energy Consumption (%)	Processing Cost (%)
Proposed Blockchain Framework	99	70	40
Traditional Authentication	90	90	70
Centralized Model	92	85	60
Hybrid Authentication Model	95	80	50

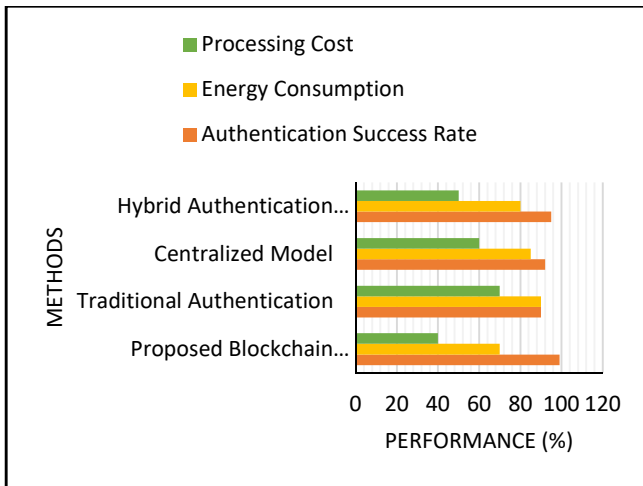


Fig 2: Performance Analysis

The blockchain structure that has been suggested obtains the greatest verification success rate, which is 99%, which reflects the structure's better dependability as shown in the table 1 and figure 2. It needs the lowest possible operating cost, which is forty percent, and it only uses seventy percent of the basic energy, which renders it very efficient. In addition to having lower rates of success, conventional methods also have higher electricity and manufacturing costs, which highlights the limits of these approaches.

Table 2: Performance analysis of each model

Technique	Communication Overhead (%)	Latency (%)	Security Level (%)
Blockchain Framework	50	30	95
Traditional Authentication	80	60	70
Centralized Model	70	55	75
Hybrid Authentication Model	60	40	85

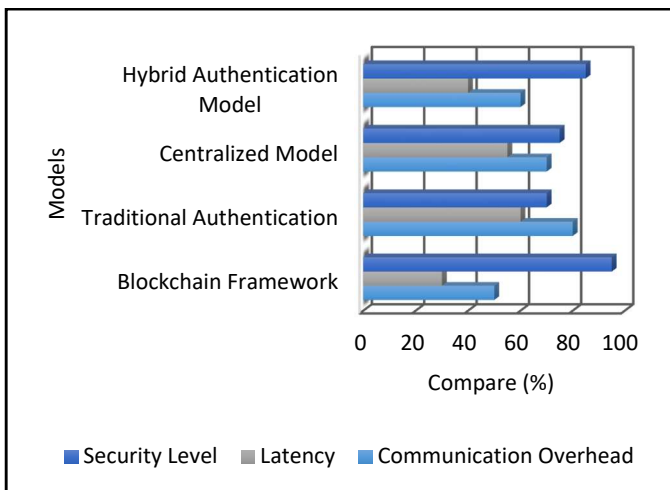


Fig 3: Performance Comparison.

With an interaction latency of just fifty percent, the architecture that has been provided ensures that data is exchanged in an effective manner. In addition to this, it has a thirty percent lower latency than previous versions, which contributes to an improved user interface. It is the structure that has been suggested that has the greatest safety level, which is 95%, proving that it provides efficient defense from attacks. Although the hybrid approach demonstrates a decent level of efficiency, conventional methods tend to trail behind in all criteria.

V. CONCLUSION

Authentication and authorization of Internet of Things (IoT) sensor devices using the blockchain system was the focus of this section. System startup, sensor device enrollment, login, safety authorization, and permission are all parts of the proposed technique. At last, we checked how well the plan worked. One way to address the issue of unreliability of third parties is by using blockchain technology as a store server to authenticate Internet of Things devices. Additional benefits of the proposed system include assured low processing, interaction, and storage expenses. Also, the inviolability of the blockchain makes it easy to find out which node did something bad. The findings prove that blockchain provides a more trustworthy means of authenticating IoT devices compared to the current methods. In order to make the current structure more scalable for IoT devices with limited resources, future studies should look at including lightweight cryptographic techniques. Further protection against potential quantum assaults may be achieved by integrating blockchain technology with post-quantum encryption techniques. Smart agreements that can react to alterations in the system and improve both efficiency and safety are another potential path. Another way to facilitate smooth communication across networks is to investigate digital currencies' interoperability among chains. Integrating detection of anomalies models powered by AI to forecast harmful activities will further improve preventative security. Adding support for portable and hybrid Internet of Things (IoT) infrastructures to the system helps enhance real-time identification in unpredictable settings.

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